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BEITI-TECH : AI Smart Home System

Research Problem

Modern households and conventional smart home systems face challenges related to security, energy efficiency, and cloud-dependent automation. These limitations can compromise user safety, increase energy consumption, and reduce system responsiveness, resulting in a less reliable and convenient smart home experience.

Key challenges include:

- Security & Safety:** Traditional access methods lack reliable identity verification, while many homes do not provide real-time detection and remote alerts for fire or gas leaks.
- Energy Inefficiency:** Manual control of lighting and appliances often leads to unnecessary power consumption.
- System Latency:** Cloud-dependent architectures introduce delays in executing local automation tasks, especially during network disruptions.

Research methodology

- Component Selection & Architecture Design:**
Selected the ESP32, ESP32-CAM, and communication protocols (ESP-NOW & Wi-Fi) to balance edge processing and cloud connectivity.
- Hardware assembly & Integration:**
Assembled the prototype by integrating environmental and security sensors (PIR, DHT22, Gas, Flame) with the ESP32 and door servo mechanism.
- Firmware & AI development:**
Programmed the ESP32 in C/C++, deployed the edge-based facial recognition model on the ESP32-CAM, and developed the Blynk IoT dashboard.
- Integrating system with architecture**

Impact / application

- Beit-Tech seamlessly converges AI, IoT, and edge computing into a unified smart home ecosystem. Its modular architecture optimizes energy efficiency, ensures proactive hazard safety, and delivers ultra-low latency automation with robust real-time monitoring.
- Application include:**
- Smart Home Security** – AI face recognition and visitor logging.
 - Safety Monitoring** – Instant fire and gas leak detection with emergency alerts.
 - Energy Management** – Automatic fan control using PWM to reduce energy consumption.
 - Remote Monitoring** – Real-time control and notifications through Blynk and Telegram.
 - Residential Automation** – Intelligent, scalable, and low-cost smart home implementation.

Key Findings

The obtained results were influenced by the project's limited **development time (10 days)** and **budget constraints**. Despite these limitations, the prototype successfully demonstrated the feasibility of an AI-powered smart home system and achieved its **primary objectives** and **validated the proposed concept**.

AI Person
Detection
accuracy

75%

Average time
response

0.3s

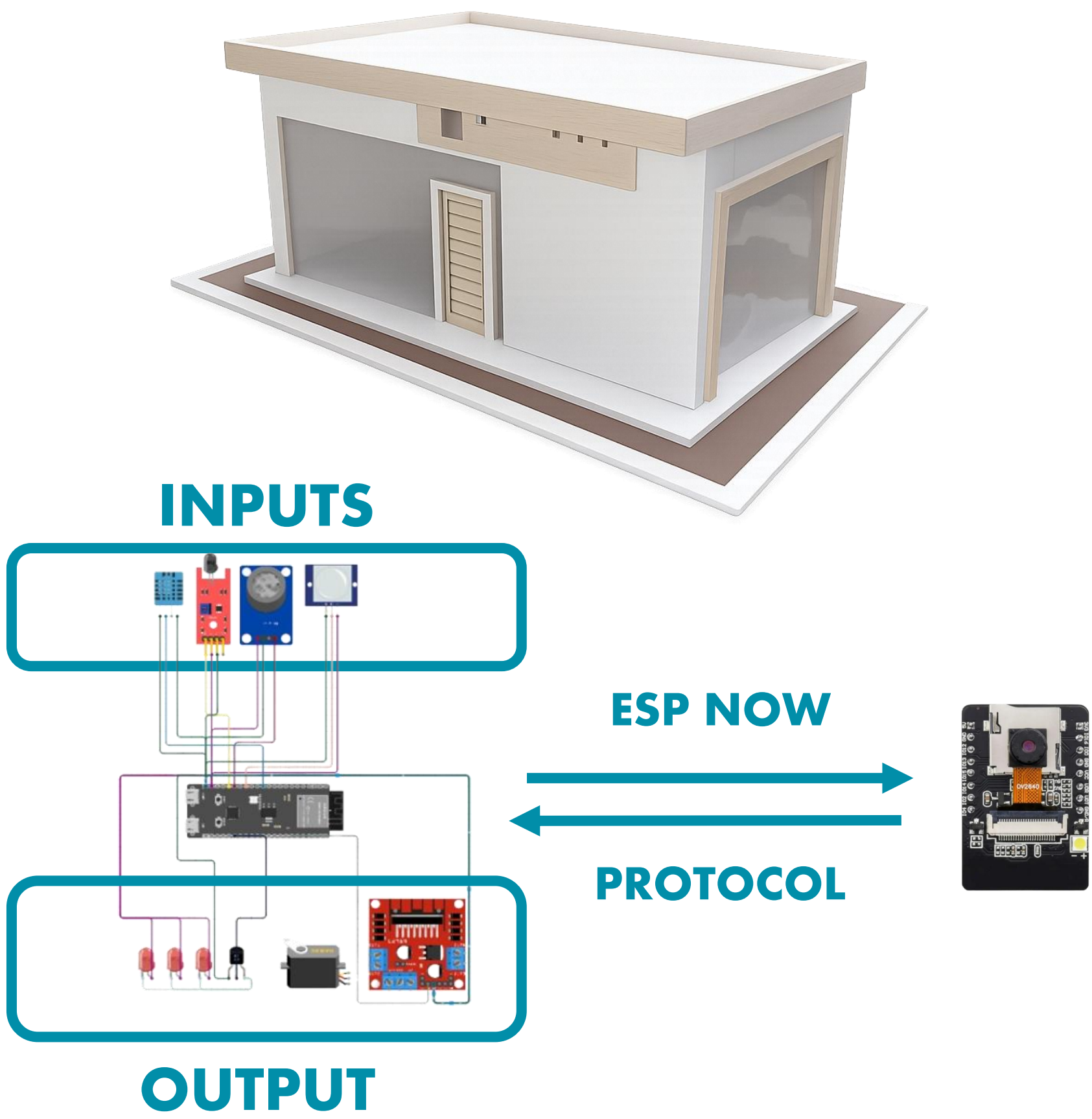
Hazard
detection
rate

98%

Door access
response
time

7s

Solution/ Prototype



System Architecture & Workflow:
The Beit-Tech prototype uses a dual-ESP32 architecture, separating AI-based access control from environmental monitoring and home automation.

1. Edge-AI Access
ESP32-CAM performs local face recognition, unlocks the door via ESP-NOW, and sends visitor photos through a Telegram Bot.

2. Smart PWM Ventilation
DHT22 controls fan speed automatically using PWM (31°C–33°C+) or allows manual control.

3. Hazard Detection
Gas and Flame sensors trigger a buzzer and send emergency alerts via Blynk.

Significant Innovation

- Unlike conventional smart home ecosystems that rely heavily on continuous cloud processing, this system introduces a **hybrid edge-cloud architecture** prioritizing speed, privacy, and offline reliability.
- Offline-First Automation:** By leveraging the ESP-NOW protocol, critical **machine-to-machine (M2M)** communications execute with near-zero latency (< 100ms) independently of Wi-Fi, ensuring core home automation survives total internet outages.
 - Edge-AI Privacy:** Facial recognition is processed locally on the vision node rather than transmitting sensitive video feeds to external third-party servers, drastically enhancing user privacy and security.
 - Algorithmic Climate Scaling:** Moves beyond rudimentary "ON/OFF" thermostats by employing intelligent PWM thermal scaling (**dynamically shifting fan RPMs between 31°C and 33°C+**) to optimize energy consumption precisely based on environmental needs.
 - Hybrid Alerting:** Seamlessly merges instant local hardware alarms with **rich cloud API integrations** (Telegram logs and Blynk notifications).

